DE–6456 **13**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2013.

DIGITAL COMPUTER FUNDAMENTALS

Time : Three hours Maximum : 100 marks

PART A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

All questions carry equal marks.

1. Explain Hexadecimal and Octal number systems. Why Binary numbers are used in digital computers?

2. (a) Subtract $(1101.11)_2$ from $(111.01)_2$.

   (b) Multiply $(1101)_2 \times (101)_2$.

3. State and prove DeMorgan's theorems.

4. Simply using K-map.

   $Y = F(A, B, C, D) = \Sigma(0, 1, 2, 4, 5, 10, 11, 14, 15)$.

5. Draw and explain full adder circuit using Nand gates.

6. What is combinational circuit and sequential circuit? Explain.

7. Draw and explain the circuit of shift registers.

8. Explain RAM and ROM.

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Explain 1's, 2's, 9's and 10's compliments with example.

10. List and explain the various Laws of Boolean algebra.

11. Simplify the function by Quine-Mc Cluskey method
$Y = F(A, B, C, D) = \Sigma(0, 5, 7, 8, 13, 15)$.

12. Explain Parallel Binary adder in detail.

13. Discuss Binary counters in detail.

14. Explain the types and uses input output units.

15. Explain computer generations and its classifications.

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DE–6457 **14**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2013

C AND DATA STRUCTURES

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the terms character set, keyword, constant and identifiers.

2. Explain branching and looping statements in C.

3. Explain the term 'Recursion' by writing a function code.

4. Write short notes on Macros and Preprocessor directives.

5. Explain file opening modes and their meaning, giving examples.

6. What is ADT? Explain.

7. Write short notes on stack and their applications.

8. Write a C program to create a linked list structure containing student data in its nodes.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Explain different data types and their memory requirements. (10)

(b) Compare break and continue statements. (5)

10. (a) Write a C program to add two 3×3 matrices. (10)

(b) Explain string handling in C. (5)

11. (a) Explain in detail about Dynamic memory allocation in C. (8)

(b) Write a C program to create and read the contents of student data file. (7)

12. Explain stack data structure and write functions for operations over a stack. (15)

13. Explain queue data structure and write functions for operations over a queue. (15)

14. Write C functions to maintain a doubly linked list. (15)

15. Write C functions to represent a binary tree using printers and traverse it. (15)

DE–6458 **15**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, MAY 2013.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. (a) Define conditional and biconditional statement.

   (b) Prove that $(P \to Q) \wedge (Q \to P) \Leftrightarrow (P \rightleftharpoons Q)$ using truth table.

2. Given $S = \{2, a\{3\}, 4\}$ and $R = \{\{a\}, 3, 4, 1\}$, indicate whether the following are true or false.

   (a) $\{a\} \in S$

   (b) $\{1, \{a\}, 3, 4\} \subseteq R$.

   (c) $\{a\} \subseteq S$

   (d) $\phi \in R$.

3. Draw Hasse diagram of $\langle \rho, (A), \subseteq \rangle$, where $A = \{a, b, c\}$ and $\rho(A)$ is the power set of A.

4. Define 1–1, onto and bijective functions.

5. Construct a function $f : \mathbf{N} \to \mathbf{N}$ such that

   (a) f is bijection but $f \neq I_n$

   (b) f is 1–1 but not onto

   (c) f is onto but not 1–1.

6. Show that the set $\mathbf{N}$ of natural numbers is a semigroup under the operation $x * y = \max\{x, y\}$. Is it monoid?

7. Prove that the kernel of a homomorphism g from a group $\langle G, * \rangle$ to $\langle H, \Delta \rangle$ is a subgroup of $\langle G, * \rangle$.

8. Define complete graph and Bipartite graph with examples.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Show that the following :

   (a) $(P \to Q) \Leftrightarrow (\neg P \vee Q)$.

   (b) $P \to (Q \to R) \Leftrightarrow P \to (\neg Q \vee R) \Leftrightarrow (P \wedge Q) \to R$

   (c) $((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee (\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)$ is a tautology.

10. Let $X = \{1, 2, \ldots 7\}$ and $R = \{\langle x, y \rangle \mid x - y$ is divisible by 3$\}$. Show that R is an equivalence relation and draw and graph of R. Determine the equivalence classes generated by the elements of X.

11. Show that the following :

   (a) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

   (b) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

   (c) $(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)$.

12. Let F_x be the set of all 1–1 onto mapping from $\{1, 2, 3\}$ onto itself. Find all the elements of F_x and find the inverse of each element.

13. Let $f : \langle G, * \rangle \to \langle H, \Delta \rangle$ be a group homomorphism. Prove that the following :

    (a) $f(e_G) = e_H$

    (b) $f(a^{-1}) = [f(a)]^{-1}$

    (c) If S is a subgroup of G then $f(S)$ is a subgroup of H.

14. State and prove the fundamental theorem of group homomorphism.

15. Let G be a simple graph with n nodes. Prove that the following statements are equivalent.

    (a) G is a tree

    (b) G has $n-1$ edges and no cycles

    (c) G has $n-1$ edges and is connected.

DE–6459 **23**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (Five Year Integrated) DEGREE EXAMINATION, MAY 2013.

OBJECT ORIENTED PROGRAMMING AND C++

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Write the applications of OOP.

2. Explain about different expressions in C++ with examples.

3. What is a class? How does a C++ structure differ from a C++ class?

4. How is a member function of a class defined?

5. Write short note on 'this' Pointer.

6. Explain Polymorphism with example.

7. What is inheritance? Explain it briefly.

8. What do you mean by static and dynamic binding? Discuss with examples.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. Explain the basic concepts of Object Oriented Programming.

10. (a) Explain about basic data types in C++.

    (b) Explain managing output with manipulators through examples.

11. (a) What is a Constructor? How do we invoke a Constructor function?

    (b) Write a program using friend function that exchange the private value of two classes.

12. (a) Write a C++ program using a Copy Constructor.

    (b) List the important functions supported by the string class.

13. (a) Explain operator overloading with suitable example.

    (b) Write a C++ program to overload — operator for subtracting two complex numbers.

14. (a) Write some of the basic rules for Virtual functions.

    (b) Write a C++ program to illustrate Virtual function.

15. Discuss on the following:

    (a) Virtual base class.

    (b) Derived and base classes.

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DE–6460 **24**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated)
DEGREE EXAMINATION, MAY 2013.

OPERATING SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Discuss the different types of OS services.

2. Explain some system calls in operating system.

3. Write short notes on operation on processes.

4. Write notes on Critical Sections.

5. What is a deadlock? Give an example.

6. What is swapping? Discuss.

7. Explain FIFO page replacement with suitable diagram.

8. Write short notes on Tree-structured directory.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. (a) Describe briefly the history of operating system.

(b) Explain about operating system structure and its system components.

10. What is Semaphore? Discuss any two applications of semaphore.

11. Explain Shortest-Job-First scheduling algorithm.

12. Describe in brief various methods for deadlock prevention.

13. Write short notes on the following:

    (a) Segmentation

    (b) Paging.

14. What is disk scheduling? Discuss any three disk space management.

15. Discuss on the following:

    (a) File servers

    (b) Protection mechanisms.

DE–6461 **25**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (Five Years Integrated)
DEGREE EXAMINATION, MAY 2013.

RELATIONAL DATABASE MANAGEMENT SYSTEM

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Write short notes on different user interfaces.

2. Discuss about the concepts of relational model.

3. Explain about data definition facilities.

4. Explain about relational database properties.

5. What are the different RDBMS packages? Explain.

6. What are the various elements of SQL language? Explain.

7. Differentiate between NULL and NOT NULL values. Give example.

8. Discuss about oracle precompiler.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. (a) What are the various features of DBMS? (5)

(b) Define schema and subschema. (5)

(c) Disadvantages of DBMS. (5)

10. (a) Explain about DML facilities. (8)

(b) What is mapping operation? (7)

11. Explain the overview of Oracle architecture.

12. (a) DML commands with example. (10)

(b) Sub queries. (5)

13. Explain about embedded SQL.

14. Discuss about :

(a) Describing table

(b) Modifying table

(c) Renaming a table

(d) Copying a table

(e) Dropping a table.

15. Explain about set operations in Oracle.

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DE–6462 **31**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, MAY 2013.

COMPUTER NETWORKS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List out and explain the various uses of computer networks.

2. Discuss about network hardware with neat diagrams.

3. Explain the various design issues of data link layer.

4. Write short notes on error detecting codes.

5. Discuss about ATM LANs.

6. Write a note on UDP.

7. Explain about cryptography.

8. Discuss about DNS.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain briefly about OSI reference model.

10. Discuss about transmission media with neat diagrams.

11. Discuss about the various elementary data link protocols.

12. Illustrate any two example data link protocols.

13. Write an essay on routing algorithms.

14. Discuss briefly about the elements of transport protocols.

15. Explain about multimedia.

DE–6463 **32**

DISTANCE EDUCATION

M.Sc. (Software Engineering)
(5 Years Integrated) DEGREE EXAMINATION,
MAY 2013.

INTERNET AND JAVA PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Describe DNS in detail.

2. How is java strongly associated with the Internet?

3. List out basic data types used in java with examples.

4. What are objects? How are they created from the class?

5. What is an applet? How do applets differ from Application Programs?

6. What is servlet? How does it differ from a java application?

7. How to create tables is HTML?

8. What is Input and Output streams explain them with illustrations.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Explain

(a) IRC

(b) Usenet Newsgroup

10. Explain features of java in detail.

11. What is a package? How do we design a package?

12. List out any five AWT controls in details.

13. Illustrate frames in HTML.

14. Distinguish between

(a) InputStream and Reader Classes

(b) OutputStream and Writer classes

15. Explain

(a) ServerSocket Class

(b) Socket Class.

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DE–6464 **33**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (Five Years Integrated)
DEGREE EXAMINATION, MAY 2013.

SOFTWARE ENGINEERING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Describe software project development concept.

2. Explain quality and productivity factor of software development.

3. Explain project estimation concept in detail.

4. Describe estimating software scope and resource in detail.

5. What are all the components required for software development?

6. Explain software specification technique.

7. Explain architectural design.

8. Explain basis path testing.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Explain
   (a) Spiral model
   (b) Linear sequential model.

10. Describe software metrics concept in detail.

11. Describe software cost estimation techniques in detail.

12. Explain
   (a) Project structure
   (b) Programming Team Structure.

13. Explain Risk Management in software development.

14. Explain problem analysis concept in detail.

15. Explain
   (a) Unit testing
   (b) Integration testing
   (c) Validation testing.

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DE–6465 **34**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated)
DEGREE EXAMINATION, MAY 2013.

COMPUTER GRAPHICS AND MULTIMEDIA

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain line drawing algorithms.

2. Discuss about video display devices.

3. Explain various transformations.

4. Explain the following:

   (a) Curve Clipping

   (b) Text Clipping

   (c) Exterior Clipping

5. Explain visualization and rendering.

6. Explain the basics of audio and its formals.

7. Discuss about various image formats.

8. Explain the presentation requirements during synchronization.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the input devices.

10. Write about pixel addressing and filled area primitives.

11. Explain line clipping methods.

12. Discuss about 3D display methods.

13. Explain in detail about computer image processing.

14. Explain about components of multimedia.

15. Explain the multimedia database system.

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DE-6466 | **35**

DISTANCE EDUCATION

M.Sc.(Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2013.

UNIX AND SHELL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions

1. Explain architecture and features of unix.

2. Discuss about unix file system.

3. Briefly discuss about Networking Commands in unix.

4. Write about unix session, redirection and pipes.

5. Explain grep operation.

6. Discuss about basic script concepts in korn shell programming.

7. Explain two special files available in korn shell.

8. Explain Direct handling commands in file management.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions

9. Write any 12 unix commands and explain in detail.

10. Explain filters in detail.

11. Write a shell Program to do the following:

    (a) To count characters, words, lines in a file

    (b) Compare two files.

12. Write about operations, string functions, mathematical function and user defined function in awk.

13. Discuss an interactive C-shell.

14. Write about file management and structures.

15. Explain the following in C-shell programming

    (a) Expression

    (b) Decision Making

    (c) Repetition

    (d) Special Parameters and Variables

    (e) Argument Validation.

DE–6658 **16**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated)
DEGREE EXAMINATION, MAY 2013.

C PROGRAMMING LAB

(Upto 2002 batch)

Time : Three hours Maximum : 100 marks

Each subdivision carries 50 marks.

ONE question should be given to each candidate by lot system.

1. (a) Write a C program to convert a given number into words for numbers from 5 to 9. Example : 5 as Five, 6 as Six, and so on.

   (b) Write a C program to calculate any nine trigonometry values using C mathematical functions.

---------- Cut here ----------

2. (a) Write a C program to find all possible roots of a quadratic equation.

   (b) Write a C program to count the number of spaces in a sentence. Example : Input : "I love my mother India"; Output: No. of spaces in the sentence = 4.

3. (a) Write a C program to Add and Subtract two matrices.

   (b) Write a C program to convert a string from Upper case to Lower case format and vice versa using C string functions.

---------- Cut here ----------

4. (a) Write a C program to Multiply two matrices.

   (b) Write a C program to find biggest of three numbers

---------- Cut here ----------

5. (a) Write a C program to print the result of the examination for 5 students using structure. The result is to be computed as more than 40% mark as "Pass", otherwise "Fail".

   (b) Write a C program to calculate simple and compound interest.

---------- Cut here ----------

6. (a) Consider the following : "A visiting professor paid Rs. 1,000 for conducting hours of lecture class per day. Thereafter for overtime Rs. 300 per hour, subject to ceiling of Rs. 600 for a day. The week days are from Sunday to Saturday. The number contact classes for a week and over time hours are up to the wisdom of the professor". Write a C program to print the weekly report as detailed below :

Week Days	No. of contact classes	Over-time hours	Claim of the day
(i) Name of the Professor : _____ & Dept. _____.			
Sun.-to-Sat.	---	---	---
		Week Total : Rs.	---
(ii) Name of the Professor : _____ & Dept. _____.			
Sun.-to-Sat.	---	---	---
		Week Total : Rs.	---
(iii)	...		---
		Week Total : Rs.	---
		Over all Total : Rs.	

(b) Write a C program to show the use of any five C string functions.

---------- Cut here ----------

7. (a) Write a C program to find the Factorial of a number using recursion.

(b) Write a C program to read the elements of an array. Use pointer to locate and display each value.

---------- Cut here ----------

8. (a) Write a C program to implement all string operations using switch statement.

(b) Write a program to find a word as palindrome or not.

DE–6659 **17**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2013.

DATA STRUCTURES LAB

Time : Three hours Maximum : 100 marks

Each subdivision carries 50 marks.

One question should be given to each candidate by lot system.

1. (a) Write a C program to copy contents of one file to another.

   (b) Write a C program to evaluate mathematical expression using stack.

---------- Cut here ----------

2. (a) Write a C program to insert and delete elements from stack.

   (b) Write a C program to sort 9 numbers in ascending order and show the values before and after sorting.

---------- Cut here ----------

3. (a) Write a C program to evaluate a mathematical expression using stack.

   (b) Write a C program to sort an array using selection sort.

4. (a) Write a menu driven program in C to find an element using Linear search method.

   (b) Write a C program to sort an array using insertion sort.

---------- Cut here ----------

5. (a) Write a C program to insert and delete elements in a queue using array.

   (b) Write a C program to copy contents of one file to another.

---------- Cut here ----------

6. (a) Write a C program for linked list implementation of queue.

   (b) Write a C program to do push and pop operations on stack.

---------- Cut here ----------

7. (a) Write a C program to insert and delete elements in a linked list.

   (b) Write a C program to sort an array using selection sort.

---------- Cut here ----------

8. (a) Write a menu driven program in C to find an element using Binary search method.

   (b) Write a C program to sort 10 numbers in descending order and show the values before and after sorting.

DE–6660 **26**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2013.

C++ — LAB

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

Break-up of marks :

Record : 10
Algorithm : 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

1. Write a C++ program to convert decimal number to any other base.

---------- Cut here ----------

2. Write a C++ program to print floyds triangle using a function.

```
1
2   3
4   5   6
7   8   9   10
11  12  13  14  15
```

3. Write a C++ program to overload arithmetic assignment operators.

---------- Cut here ----------

4. Write a C++ program using operator function as Friend function.

---------- Cut here ----------

5. Write a C++ program to implement the concept of destructor with an example.

---------- Cut here ----------

6. Write a C++ program to implement the concept of multilevel inheritance with an example.

---------- Cut here ----------

7. Write a C++ program to illustrate a virtual function.

---------- Cut here ----------

8. Write a C++ program to insert and delete elements in a linear list.

---------- Cut here ----------

9. Using pointers create a class and write a C++ program to sort n numbers in Ascending and Descending order.

---------- Cut here ----------

10. Write a C++ program to arrange the names in alphabetical order using pointers.

DE–6661 **27**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2013.

RELATIONAL DATABASE MANAGEMENT SYSTEM — LAB

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

1. The NSS unit of a college maintains a blood denors record information in a database. The database contains the following fields :

   Donor – name, Donor – no, age, address, Date-of-birth, Blood-group (A, B, AB, etc), Sex.

   Write a PL/SQL procedure to display the number, name and address of the donors for the following categories :

   (a) Female donors with "AB" blood group who have age between 20 and 23.

   (b) Blood donors who have "O +" blood group.

   (c) Blood donors with age between 18 to 20.

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2. Write a PL/SQL procedure to prepare student's mark list using cursor.

   The fields to be created are : Reg-no, name, mark 1, mark 2, mark 3, total, average, grade, result.

   if average > = 60 – Grade A

   if average 40 to 59 – Grade B

   if average > = 35 and loss than 40 – grade C otherwise, "Fail".

3. (a) Create an EMPLOYEE table to calculate the salary of all employees (atleast 10 records) with the following fields : Employee_No, Employee_Name, Age, Date-of-joining, Sex, Basic, DA, HRA, PF, NET.

DA = 50% of Basic

HRA = 20% of Basic

PF = 10% of Basic

NET = BP + DA + HRA – PF.

Write a PL/SQL procedure to print the employee details in a neat format using cursor.

(b) Write Queries for the following :

(i) Display the employee details in the descending order of their names.

(ii) Display the employees who get more than Rs. 30,000 as basic pay.

(iii) Display the female employees whose age is in between 46 to 56.

---------- Cut here ----------

4. A railway reservation database contains the reservation table and personal table. Reservation table contains following fields :

Train – no, Train – name, Class, Passenger – name, Coach – no, Seat – no.

Personal table contains following fields : Passenger-name, Sex, age,

Write a program to prepare the following report.

(a) List the passengers names with seat number according to class and Train-no.

(b) Total number of female passengers in a particular train who have age between 40 to 50.

(c) List out the male passengers in the age above 60.

5. Write a PL/SQL to prepare Electric bill using cursor.

The table contains the following fields :

EB-no, customer – name, previous – reading,

Current – reading, actual – reading, amount, total.

Actual reading = current – reading – previous – reading.

if actual reading < = 100, amount = actual – reading * 10

if actual reading > 100 and < = 200, amount = actual – reading * 15

if actual reading > 200 and < = 400, amount = actual – reading * 18

if actual reading > 400, amount = actual – reading * 25.

Print the output in the following format :

EB-no customer-name, previous-reading current – reading amount total.

DE–6662 **36**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2013.

INTERNET AND JAVA – LAB

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

Each subdivision carries 50 marks.

1. (a) Write a Java script to create a window by using the confirm message.

   (b) Design and publish a web page for a college.

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2. (a) Write a VB script to do the following:

   (i) Check the given password

   (ii) Change the existing password.

   (b) Develop a web page for job recruitment agency in an IT industry.

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3. (a) Write a VB script to do the following :

   (i) Display the current date

   (ii) Find the difference between the two dates

   (iii) Find the age of a person by providing date of birth For all use the date in dd/mm/yy format.

   (b) Develop a single page advertisement for a shop to be opened recently.

4. (a) Write a VB script to create a calendar for given month and year.

   (b) Create a web page in the format of front page of a news paper using Text link Align the text with colors.

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5. (a) Write a Java program using applet to display the dialogue menu in applet.

   (b) Using atleast 20 HTML tags create a screen with a string "WEB Design".

DE–6663 **37**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years integrated) DEGREE EXAMINATION, MAY 2013.

UNIX AND SHELL PROGRAMMING – LAB

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

Each subdivision carries 50 marks.

1. (a) Write a Menu Driven shell program to copy, edit, rename, delete a file.

   (b) Write a Menu Driven shell program to perform the following tasks :

   (i) Enter the sentence in a file

   (ii) Search a given whole word in an existing file

   (iii) Quit.

---------- Cut here ----------

2. (a) A Hospital maintains patient details. Write a shell program to list the following :

   (i) Patients in a blood groupwise.

   (ii) Patients in age between 20 to 30.

   (b) Write a program for four-function calculator by using yacc command.

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3. (a) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found, display "Pattern found" else display "Error message". Also check if right number of arguments are entered.

   (b) Write and execute a shell program to print monthly payslips for employees in an organization. Assume your own data and generate accordingly.

4. (a) Write and execute a shell program to generate leave status report of an organization.

(b) Write and execute a shell program to send mail to groups of users by extracting their Ids front/etc/group file.

---------- Cut here ----------

5. (a) Write a shell program to prepare electric bill for domestic consumers.

For first 100 units – Rs.0.75/unit

For next 100 units – Rs. 1.50/unit

Above 200 units – Rs. 3.00/unit

Prepare the Electricity bill in the following format :

Customer no ————

Customer name ————

Previous reading ————

Current reading ————

Units consumed ————

Charge ————

(b) Write and execute a shell program to do the following :

(i) Convert given string in center of given row in BOLD, normal or reverse video.

(ii) Write given string at given row, column in BOLD, Normal and reverse video.

(iii) Search a given string using awk with various options.

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